

# RÚBEN ARAÚJO

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 Portuguese (EU)

## RESEARCH EXPERIENCE (Biomedical / Clinical)

**Researcher – Instituto Superior de Engenharia de Lisboa (ISEL)** Oct 2017 – Present | Lisbon, Portugal

- Led and supported biomedical and clinical research activities focused on biospecimen handling, sample preparation, protocol optimization, FTIR metabolomics, and targeted proteomics assays. Supported MSc students and PhD colleagues across projects involving transplantation rejection, infection, mortality prognosis, embryo viability, genotoxicity, and bioactive compounds.

**Researcher – Comprehensive Health Research Center (CHRC)** Nov 2021 – Present | Lisbon, Portugal

- Integrated multidisciplinary clinical and biomedical research groups, contributing to biomarker-focused studies and collaborative scientific activities.

**Studentship – Fundação para a Ciência e a Tecnologia (FCT)** Nov 2021 – Nov 2025 | Lisbon, Portugal

- Doctoral research focused on ICU-based biomarker discovery and clinical outcome stratification.

**Principal Investigator – Instituto Português da Face (IPF)** 2020 | Lisbon, Portugal

- Led clinical research on TMJ tissue regeneration and 3D surgical planning, including sample/data processing, statistical analysis, and scientific publication.

## BIOMEDICAL OMICS & BIOMARKER RESEARCH

- Blood-based biomarker** research in critical care, infection diagnosis, and mortality prognosis.
- FTIR-based molecular fingerprinting / metabolomics** applied to clinical serum samples.
- Targeted proteomics** through multiplex cytokine assays.
- Machine-learning**-supported biomedical classification and clinical data interpretation.
- Translational research** involving ICU cohorts, clinical metadata, and biomarker-driven outcome stratification.

## RESEARCH PROJECTS

- Participation in **8 funded research projects** as Researcher, Principal Researcher, and Founder, with a **combined funding across projects** of ≈1 million Euros.
- Domains:** clinical metabolomics, biomedical devices, infection diagnostics, critical care, TMJ regenerative solutions, nephrology, machine learning applied to clinical medicine.
- Institutions involved:** NMS, ISEL, CHULC, IST, IPST, IPL, and Fundação Champalimaud.
- Experience with FCT, Portugal 2020 / FEDER, and multidisciplinary biomedical research frameworks.

## PUBLICATIONS & SCIENTIFIC OUTPUT

- 30+ peer-reviewed publications**, including **9 first-author** articles.
- Quartiles:** 13 Q1, 13 Q2, 2 Q3, 1 Q4.
- Topics:** infectious diseases, critical care mortality, nephrology, transplantation rejection, genotoxicity, extracellular vesicles, embryo viability, and bioactive food compounds.

## MEDICAL RESEARCH INNOVATION

- National patent application:** PT120761 (2025), Infection Diagnosis.
- International patent contribution:** PCT/PT2025/050006, Allograft Rejection Prognosis.
- Experience supporting patent documentation, prior-art positioning, and research-to-application discussions in biomedical projects.

## EDUCATION

**PhD in Biomedicine – NOVA Medical School (NMS)** 2021 – 2026 | Lisbon, Portugal

**MSc in Biomedical Engineering – ISEL** 2017 – 2019 | Lisbon, Portugal

**MSc in Mechanical Engineering (Maintenance & Production) – ISEL** 2007 – 2011 | Lisbon, Portugal

**BSc in Mechanical Engineering – ISEL** 2002 – 2007 | Lisbon, Portugal

## SCIENTIFIC INFRASTRUCTURE & RESEARCH MANAGEMENT

|                                        |
|----------------------------------------|
| Scientific Project Management          |
| Research Infrastructure Implementation |
| Translational Biomedical Research      |
| Omics-Based Biomarker Development      |
| Clinical Research Coordination         |
| EU & National Funding Frameworks       |
| Scientific Reporting & Documentation   |
| Multidisciplinary Team Interface       |

## ANALYTICAL & TECHNICAL TOOLS

|                                      |
|--------------------------------------|
| Python / R / SQL / Power BI/ Tableau |
| Reporting & Documentation            |

## LANGUAGES

- Portuguese** – Native
- English** – C1 (IELTS + Cambridge)
- Norwegian** – B1 (Aftenskolen)
- Spanish** – B1 (Instituto Cervantes)
- French** – Basic

## ACADEMIC LEADERSHIP

- Erasmus Student Coordinator (ISEL)** | 2018-2019
- Founder, EMBS Student Branch (ISEL)** | 2018
- Chair, IEEE Student Branch (ISEL)** | 2017-2018
- Class Representative Bio. Eng. (ISEL)** | 2017-2019

## AWARDS & SCHOLARSHIPS

- FCT PhD Studentship** | 2021-2025
- First in Class (PhD)** | 2023
- Research Scholarship (ISEL)** | 2021
- Merit Scholarship (Ministry of Education)** | 2021
- First in Class (MSc Bio. Eng)** | 2020

## ENGINEERING EXPERIENCE

- CEO – Memory Particle (Portugal)** | 2017-2018
- Lead Mechanical – Brøvig (Norway)** | 2016-2017
- Life Cycle Engineer – MH Wirth (Norway)** | 2014-2015
- Project Engineer – Aker Solutions (Norway)** | 2012-2014
- Mechanical Design – Siemens (Portugal)** | 2011-2012

*Note: Senior engineering experience in large-scale, multidisciplinary and multinational industrial projects, involving coordination across diverse technical teams, communication with multiple stakeholder groups, and interface with executive-level decision makers.*

All supporting documentation can be consulted and downloaded @ <https://rubenaraujo.org/cv/>